

Class test.

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(a)

Define Data Mining:-

Data Mining is the process of extracting useful patterns, relationships and knowledge from large amounts of data.

Two Basic Data Mining Tasks:-

1) Classification - Assigns data to predefined classes.

Example:- Spam or Not Spam Email.

2) Clustering - Groups similar data without predefined classes.

Example:- Grouping customers based on purchasing behaviour.

Steps for KDD (Knowledge Discovery in Databases):-

- 1) Data Cleaning
- 2) Data Integration
- 3) Data Selection
- 4) Data Transformation
- 5) Data Mining
- 6) Pattern Evaluation
- 7) Knowledge Presentation

Role of Data Processing:-

- Improves data Quality
- Removes missing values and noise
- Reduces Redundancy.

b) Data cleaning It is the process of detecting inaccurate, incomplete or inconsistent data.

Two common Data Quality Issues.

- 1) Missing values
- 2) Noisy Data.

Methods to Handle Missing values

- Ignore the record
- Replace with median/mode/mean
- predict missing values using machine learning

Example:

$$\text{Marks} = 70, 75, -, 80, 85$$

$$\text{Mean} = 77.5$$

Replace Missing value with 77.5

Methods to Handle Noisy Data.

- Binning
- Regression
- clustering
- Smoothing

example Mark: 50, 51, 49, 120

120 is an outlier and can be removed or corrected.

a)

Data Integration:-

Data Integration combines data from multiple sources into a single, consistent dataset.

Two challenges:

1) Data Redundancy

2) Data Inconsistency

Handling Redundancy:

→ Remove duplicate records

→ use normalization.

→ correlation analysis.

Handling Inconsistency:-

→ Standardize format

→ Resolve naming conflicts

→ validate data.

b)

Covariance Analysis

Covariance Analysis measures how two variables change together.

Population Covariance:-

$$\text{Cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample:-

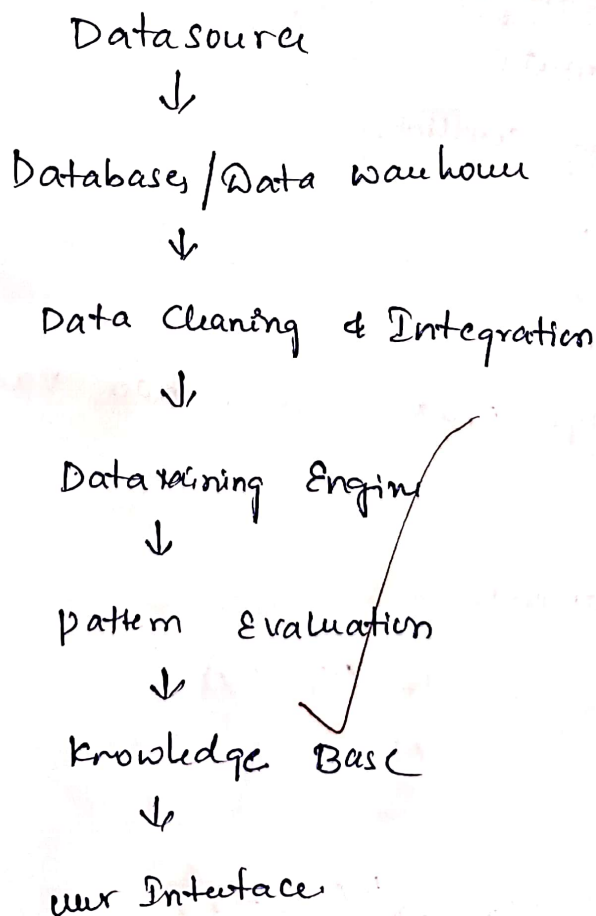
$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Interpretation:

- positive $\rightarrow$ variables increase together
- Negative $\rightarrow$ One increases while the other decreases.
- zero $\rightarrow$ no linear relationship.

3) a) After calculating the covariance for given stock values the result is 5, which is positive. The A and B move in the same direction. Therefore, they are affected by similar industry trends and generally rise or fall together.

b) Architecture of a Data Mining System:-



(a) Finding Covariance

$$X = \{4, 5, 6, 7, 9\} ; Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = 6.2$$

$$\text{Mean}(Y) = 5.8$$

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1} = \frac{-1.8}{4} = -0.45$$

$$\therefore \text{Covariance} = -0.45$$

(b) Data - preprocessing:

Major steps:

- Data Integration
- Data Cleaning
- Data Transformation
- Data Reduction
- Data Discretization.

preprocessing pipeline:-

- Handle missing values
- Remove Outliers
- Standardize data formats
- Transform/normalize data
- Reduce unnecessary data
- Apply discretization (if needed).

5a) Local Smoothing using equal - frequency Binning.

Given data = { 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 }

Divide into 3 equal - frequency (9 values each)

Bin 1:- 13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18

smoothed : 18, 18, 18, 18, 18, 18, 18, 18, 18.

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Mean = 25

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Mean = 44.

Data Discretization:-

Data Discretization is the process of converting continuous data into intervals.

Importance

- Reduces data size
- Removes noise
- Improves Analysis & Mining Efficiency

Ex: 0-20, 21-40, 41-60

→ Each bin contains the same number of data values.